

Phases 3, 4 and 5: what was built

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All three phases are on the server. This is what each one does, what changed after your testing, and the few things the game itself will not allow.

1 Phase 3: the bottom bar

The five buttons work. The server decides whether you may press one, holds the car still for as long as the job takes, and adds the penalty. The game does the work.

Button	Hold	Penalty	What it does
Reposition	5s	60s	Back on its wheels where it stands. The damage stays.
Spare tire	30s	30s	Air back into the flat tires, in place. See section 2.
Repair	60s	30s	The whole car rebuilt, in place.
Fuel +25%	20s	none	A quarter of a tank.
Lights	none	none	Headlights, fog, hazards, lightbar, siren, horn, flash.

The penalty lands when the hold starts, not when it ends, so quitting half way through buys nothing. A job the game cannot do gives the seconds back. Outside a run everything is instant and free. Every number lives on the server and can change without anyone reinstalling.

After your testing, three behaviours were corrected: nothing teleports any more. Reposition rights the car where it is and repairs nothing. Repair fixes the car where it is. Both go through the game's own safe placement, so a car against a wall is set down beside it rather than inside it.

2 The spare tire, honestly

You had seen a tire fixed without a full reset, and you were right. The game cannot mend a tire, but it can put air back in, because that is what its own onboard air compressors do. So Spare tire now refills every flat tire in place and clears the flat, touching nothing else.

The limit: a tire that is shredded or on a broken wheel will not come back from air. In that case the button falls back to the full repair and tells you so. Nearly every racing flat is the recoverable kind.

3 Phase 4: races you make and join

The touchy motion start is gone. A race is a thing you make, and others join.

1. Pick a course, laps and mode, and make the race **public** or **invite only**, choosing people from the player list.
2. Everyone who joins is placed on the grid. Nothing is armed. The start line does nothing, however many times it is crossed.
3. The host presses **Start Race**. Only then does the line go live.
4. Each driver's clock starts when they cross it, so the timing stays exact for everyone even if they roll off a second apart.

Only the host can start it. A race that has started refuses late joiners. If the host leaves before starting, the race passes to whoever joined first rather than dying. One race per course, different courses at the same time are fine, and solo practice works exactly as before.

4 Phase 4: the pit and the speed zones

A checkpoint can be marked as a **pit** when a course is built. It is not part of the lap, it is drawn in its own colour, and while you are inside it:

- Fuel fills to 100% instead of a quarter.
- Repair and Spare tire add nothing to your race time.
- The holds still run in full, and the clock is still going. A stop costs the time it takes and nothing on top.

Speed zones are the 37 mph idea, between any two checkpoints you pick. A zone is a start gate, an end gate and a limit, and a course can have as many as needed. The server reads the speeds it already collects, so nothing is trusted to a driver's own game. Going over draws a warning first; the 30 second penalty lands only after three seconds over, so one bump over a crest is free, and it will not charge twice until you have dropped back under.

5 Phase 5: the record books

Every course keeps a book per mode. The Records button shows them.

- Each driver's best corrected run, fastest first, with when it was set.
- The fastest clean lap anyone has driven, held separately.
- Your own position when you are outside the top ten.
- Controller and wheel are separate books, as agreed in phase 2.

The moment a race finishes, the run goes into the books before the results go out, so the results screen hangs the badge on the row it belongs to: **course record** or **personal best**. The books live on disk and survive restarts.

What never makes the books: a run the server marked as suspect, because its clock could not be vouched for, and any lap with a cut checkpoint in it, which was already refused a best lap by the results.

6 Fixed along the way

The recordings you sent found real bugs, and each one led somewhere useful.

What you saw	What it was
Checkpoints only counted near the middle of the road.	A gate is centred on wherever the capture car drove. Gates now measure the track to each side and slide to span the whole lane.
Some checkpoints floated and cars passed under them.	The volume was centred on its base. It now stands from just below the ground to its full height.

What you saw	What it was
Missing one checkpoint froze the ones after it.	Several causes, all gone: the volume was a slot along the road instead of a gate across it, and a run that skipped too many gates was refused forever.
The clock read 4:60.0.	The minutes were split off before the rounding.
Repair teleported the car across the map.	A bare engine reset means "back to spawn". Both movers now place the car exactly where it stands.
The clock jumped backwards thirty seconds.	A checkpoint charged as cut had been reached after all, and the penalty came back off. Correct, but silent; it now says so on screen.
The light menu would not close, and popups sat around.	Everything closes on a click elsewhere or Escape, and messages run a ten second line and go.

7 The numbers

Suite	Tests
Gate geometry	12
Course capture and shape	31
Phase 1, identity	91
Phase 2, racing	131
Phase 3, the bottom bar	50
Phase 4, lobby, pit, zones	56
Phase 5, the books	31
Results	103
The clock and lap bar	26
Total	531

Plus 146 interface checks and 10 key bindings that fail the build when the screen and the code stop agreeing. The books, the bar and the lobby were also tested by breaking them on purpose: charging at the wrong end, letting marked runs into the books, losing the store. Every sabotage was caught, which is what says the numbers above mean something.

8 What the game will not allow

Written down so nobody goes looking for a setting that does not exist.

- One tire cannot be mended by itself. Air can be put back, which covers punctures; a shredded tire needs the full repair.
- A car cannot be frozen for one player only in a way other players do not see. The hold freezes the car for everyone, which is correct for a race anyway.

9 Still open

- The tire shop sound for the spare. It needs a sound file from you, or I use one of the game's.
- The look of the interface, which you asked about. It is a separate job and can start whenever you like.
- CoPilot and Challenges are phase 6.

All of the above is deployed. The books start empty, so the first finished race on each course sets its first record.